

TABLEAU

Module 4 - Sorting, Filtering & Top N in Tableau

What You Will Learn in This Session

Topic 1	Sorting — 4 Types with real-life examples and step-by-step guide
Topic 2	Filtering — 5 Types explained simply with analogies and practice
Topic 3	Top N Filters — Built-in method + Dynamic Parameter method
Bonus	Filter Order of Operations — the most misunderstood concept for beginners

Beginner-Friendly • Real-Life Analogies • Practical Dataset Throughout

Our Practice Dataset — Module 4

We will use ONE consistent dataset throughout all topics in this module. This is sales data from a retail electronics shop across 5 Indian cities. Every sorting, filter, and Top N example will be based on this same data — so you can follow along easily.

Row	Product	Category	City	Sales (₹)	Profit (₹)	Qty	Month
1	Laptop	Electronics	Mumbai	62,000	14,000	1	January
2	Laptop	Electronics	Delhi	55,000	11,000	1	January
3	Mobile Phone	Electronics	Mumbai	36,000	8,000	2	January
4	Mobile Phone	Electronics	Bangalore	32,000	7,200	2	February
5	Tablet	Electronics	Chennai	25,000	5,500	1	February
6	Tablet	Electronics	Delhi	22,000	4,800	1	February
7	Headphones	Accessories	Mumbai	10,500	3,200	3	March
8	Headphones	Accessories	Pune	8,400	2,600	2	March
9	Keyboard	Accessories	Delhi	6,200	1,800	2	March
10	Keyboard	Accessories	Chennai	4,800	1,200	3	March

How to Read This Dataset

Each row is one sale. We have 2 Categories (Electronics, Accessories), 5 Products, 5 Cities, and 3 months of data. Sales range from ₹4,800 to ₹62,000. We will sort, filter, and rank this data throughout the module.

Topic 1 — Sorting in Tableau

Sorting means arranging your data in a specific order — highest to lowest, A to Z, or in a custom order you define. Without sorting, a chart is just a jumble. With sorting, patterns jump out immediately.

Real-life analogy

Think of a cricket scorecard. You could list batsmen in batting order (1 to 11), or sort them by runs scored (highest first). The data is the same — but sorting changes what story the scorecard tells. In Tableau, sorting tells your chart's story.

1.1 Why Sorting Matters

Without sorting, a bar chart of Sales by Product might show: Keyboard, Tablet, Headphones, Laptop, Mobile Phone — in random order. Nobody can quickly see which product sells the most.

With sorting (Sales descending): Laptop → Mobile Phone → Tablet → Headphones → Keyboard. Now the story is instant and clear.

1.2 The 4 Types of Sorting

1. Alphabetical Sort

2. Computed Sort

3. Manual Sort

4. Nested Sort

Type 1 — Alphabetical Sort

Sorts text fields from A→Z or Z→A. This is the simplest sort — just like sorting names in a phone book or a dictionary.

Real-life example: *You have a list of 5 cities — Mumbai, Delhi, Bangalore, Chennai, Pune. Alphabetical sort (A→Z) gives: Bangalore, Chennai, Delhi, Mumbai, Pune.*

Result on our dataset — City sorted A→Z:

Sort Position	City	Total Sales (₹)
1st	Bangalore	32,000
2nd	Chennai	29,800
3rd	Delhi	83,200
4th	Mumbai	1,08,500
5th	Pune	8,400

How to do it in Tableau:

1	Build a bar chart — City on Rows, SUM(Sales) on Columns
2	Click the Sort icon on the City axis (the small A→Z icon that appears when you hover)
3	Click once for A→Z (ascending alphabetical)
4	Click again for Z→A (descending alphabetical)
5	Or right-click the City pill on Rows → Sort → Alphabetic → Ascending / Descending

When to use Alphabetical Sort?

Use it when there is no natural ranking and you just want consistent, easy-to-find labels.

Example: A dropdown list of product names, a reference table, a report that users will read top-to-bottom looking for a specific city.

Type 2 — Computed Sort (Sort by a Measure)

Sorts by a calculated number — like sorting products from highest sales to lowest. This is the most useful sort for business dashboards. You always want to see your best performer at the top.

Real-life example: *An Amazon bestseller list is a Computed Sort — products are ranked by number of units sold. The top-selling item is always at the top, and the list updates as sales change.*

Result — Product sorted by Sales descending (highest first):

Rank	Product	Total Sales (₹)	Insight
1	Laptop	1,17,000	Best seller — nearly half of all revenue
2	Mobile Phone	68,000	Strong second
3	Tablet	47,000	Solid mid-range
4	Headphones	18,900	Low revenue but high quantity
5	Keyboard	11,000	Lowest — review pricing or promotion

How to do it in Tableau:

1	Build chart — Product on Rows, SUM(Sales) on Columns
2	Method A (Quick): Click the Sort Descending button in the toolbar (looks like bars going down with an arrow)

3	Method B (Full control): Right-click the Product pill on the Rows shelf → Sort
4	In the Sort dialog: Sort Order = Descending, Sort By = Field, Field Name = Sales, Aggregation = Sum
5	Click OK — bars automatically rearrange from highest to lowest
6	To reverse: choose Ascending in the same dialog

Type 3 — Manual Sort

You drag items into YOUR preferred custom order — regardless of alphabetical or numeric logic. Useful when the correct order is based on business knowledge, not just numbers.

Real-life example: A menu at a restaurant lists Starters → Main Course → Desserts. This is not alphabetical and not by price. Someone manually decided the order because that is the logical flow of a meal.

Another example — Months of the year. Alphabetically: April, August, December, February... which is useless! You want: January, February, March... in calendar order. Manual Sort (or Sort by the date field) fixes this.

How to do it in Tableau:

1	Right-click the dimension pill (e.g. Month) on the Rows or Columns shelf
2	Click Sort → Manual
3	A list of all values appears with up/down arrows on the side
4	Drag items or use the arrows to arrange them in your desired order
5	Example: drag 'January' to top, 'February' second, 'March' third
6	Click OK — the chart updates to your custom order

When to use Manual Sort?

Use it for things with a natural human order that numbers cannot capture: months of the year, days of the week, shirt sizes (S→M→L→XL), stages of a process (Lead→Prospect→Customer), or priority levels (High→Medium→Low).

Type 4 — Nested Sort

Nested Sort is sorting within groups. When you have multiple dimensions on the same axis, Nested Sort makes sure the sort applies WITHIN each group separately — not just across the whole chart.

Real-life example: *In a school report, each class is sorted separately by marks — not all students across all classes together. Class A toppers are shown first within Class A, Class B toppers first within Class B. That is a Nested Sort.*

Example on our dataset — Category and Product on Rows, sorted by Sales:

Category	Product	Sales (₹)	Nested Sort Result
Accessories	Headphones	18,900	← #1 in Accessories
Accessories	Keyboard	11,000	← #2 in Accessories
Electronics	Laptop	1,17,000	← #1 in Electronics
Electronics	Mobile Phone	68,000	← #2 in Electronics
Electronics	Tablet	47,000	← #3 in Electronics

Each category has its own internal ranking. Laptop is #1 in Electronics. Headphones is #1 in Accessories.

How to do it in Tableau:

1	Build chart with TWO dimensions on Rows: e.g. Category and Product (Category first, Product second)
2	Add SUM(Sales) on Columns
3	Right-click the INNER dimension pill (Product) → Sort
4	Sort Order = Descending, Sort By = Field, Field Name = Sales, Aggregation = Sum
5	Check the box 'Sort within panes' — this is what makes it a NESTED sort
6	Click OK — each category now has its own internal sort order

Without Nested Sort vs With Nested Sort

WITHOUT: Products sorted globally — Laptop, Mobile Phone, Tablet, Headphones, Keyboard across all categories mixed together.

WITH Nested Sort: Each Category shows its own top product first. Much easier to compare performance within a group.

Topic 2 — Filtering in Tableau

Filtering means showing ONLY the data you want to see, and hiding everything else. Instead of looking at all 10 rows every time, a filter lets you focus — show only Mumbai sales, show only Electronics, show only months where profit > ₹5,000.

Real-life analogy

Think of using Swiggy or Zomato to order food. You use filters to narrow down: Cuisine = Chinese, Rating = 4 stars+, Delivery time = under 30 minutes. The full list has 500 restaurants — the filter shows you only the 8 that match. You have not deleted the other restaurants; you have just hidden them temporarily.

2.1 The Filter Order of Operations

This is the most misunderstood concept for beginners. Tableau runs filters in a specific sequence. The filter that runs first affects all the filters that come after it. Understanding this prevents many confusing bugs.

Tableau Filter Order — from FIRST to LAST:

Order	Filter Type	What It Does
1st	Extract Filter	<i>Runs at data source level — filters the raw data BEFORE it even enters Tableau</i>
2nd	Data Source Filter	<i>Limits which data comes from the source — like a permanent filter on the connection</i>
3rd	Context Filter	<i>Sets the stage — all other filters run WITHIN the context this defines</i>
4th	Dimension Filter	<i>Filters by category fields — e.g. show only Electronics</i>
5th	Measure Filter	<i>Filters by numbers — e.g. show only Sales > 20,000</i>
6th	Table Calc Filter	<i>Filters AFTER table calculations are complete — rank, running total, etc.</i>

Why Does Order Matter?

If you add a Context Filter for 'Electronics only', Tableau re-runs everything with only Electronics rows. A Top 5 filter that comes AFTER will then give Top 5 Electronics — not Top 5 of everything. Without Context Filter, Top 5 might pull from all categories, then show only Electronics — giving you fewer than 5 results. More on this in Topic 3.

2.2 The 4 Types of Filters

1. Dimension Filter	2. Measure Filter	3. Date Filter	4. Context Filter
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Filter Type 1 — Dimension Filter

Filters based on CATEGORY fields — text, names, labels. You pick which values to INCLUDE or EXCLUDE. Most common filter type for beginners.

Real-life example: *On LinkedIn Jobs, you filter by Location = 'Mumbai' and Job Type = 'Full-time'. You are filtering a text/category field — you only see jobs from Mumbai, not Bangalore or Delhi.*

Example on our dataset — Filter: Category = Electronics only

All Data (No Filter)	After Dimension Filter: Electronics Only
10 rows total	5 rows (only Electronics)
Shows Laptops, Mobiles, Tablets, Headphones, Keyboards	Shows only Laptops, Mobiles, Tablets
Accessories rows visible	Accessories rows hidden (not deleted)

How to do it in Tableau:

1	In the Data panel, find the 'Category' field
2	Drag 'Category' to the Filters shelf (the shelf above the Marks card)
3	A dialog opens — 'Filter [Category]'. You see a list of all values: Accessories, Electronics
4	Check the boxes for values you WANT to see — e.g. check 'Electronics', uncheck 'Accessories'
5	Click OK — your chart now shows only Electronics data
6	To show the filter as a dropdown on the chart: right-click 'Category' on the Filters shelf → Show Filter
7	A dropdown appears on the right side of your view — users can now change the filter themselves

Dimension Filter Options — what the dialog gives you:

- **General tab:** Check/uncheck specific values from a list
- **Wildcard tab:** Filter by partial text — e.g. contains 'Mobile' finds 'Mobile Phone'
- **Condition tab:** Filter based on a formula — e.g. SUM(Sales) > 30,000
- **Top tab:** Show Top N or Bottom N values — we cover this in detail in Topic 3!

Filter Type 2 — Measure Filter

Filters based on NUMBERS. You set a range or condition — show only rows where Sales is above a threshold, or Profit is between two values.

Real-life example: *On MagicBricks property search, you set Price Range = ₹50 lakh to ₹1 crore. You are filtering a numeric field. Properties outside that price range are hidden.*

Example — Filter: Show only sales where Sales (₹) >= 20,000

Product	City	Sales (₹)	Passes Filter?
Laptop	Mumbai	62,000	✓ Yes — above 20,000
Laptop	Delhi	55,000	✓ Yes — above 20,000
Mobile Phone	Mumbai	36,000	✓ Yes — above 20,000
Mobile Phone	Bangalore	32,000	✓ Yes — above 20,000
Tablet	Chennai	25,000	✓ Yes — above 20,000
Tablet	Delhi	22,000	✓ Yes — above 20,000
Headphones	Mumbai	10,500	✗ No — below 20,000
Keyboard	Delhi	6,200	✗ No — below 20,000

How to do it in Tableau:

1	In the Data panel, find the measure you want to filter — e.g. 'Sales (₹)'
2	Drag 'Sales (₹)' to the Filters shelf
3	A dialog opens — choose how to filter: → 'Sum of Sales' (aggregate — most common for charts) → Choose 'Range of Values' to set min and max
4	Click Next → A slider appears. Set minimum = 20,000, maximum = leave as max
5	Click OK — only rows with Sales >= ₹20,000 are shown (6 rows remain)
6	You can also choose 'At least', 'At most', or specific values in the same dialog

Filter Type 3 — Date Filter

Filters based on DATE fields. Tableau gives you very smart options — you can filter by specific dates, relative dates (last 7 days), or by date parts (only Tuesdays, only Q3).

Real-life example: *Your bank statement app shows you transactions for 'This Month', 'Last 3 Months', or a custom date range. That is a Date Filter — you are only seeing data within a time window.*

Date Filter has 4 powerful options:

Option	What It Does	Example
Relative Date	Filter based on today's date dynamically — updates every day automatically	<i>Last 30 days, This quarter, Last month — always current</i>
Range of Dates	Set a fixed start and end date — like a date range picker	<i>01-Jan-2024 to 31-Mar-2024</i>
Date Part	Filter only specific parts — year, quarter, month, day, weekday	<i>Show only data from 'March' across all years</i>
Individual Dates	Pick specific exact dates from a calendar	<i>Only 05-Jan-2024 and 18-Jan-2024</i>

Example — Filter: Show only data from January and February (filter out March):

1	Drag 'Month' (or 'Order Date') to the Filters shelf
2	Tableau asks what to filter — choose 'Month / Year' for date part filtering
3	A calendar-style dialog opens — check January and February, uncheck March
4	Click OK — your chart now shows only Jan and Feb data (8 rows, March rows hidden)
5	To make this a 'Relative Date' filter instead: drag the date field again, choose 'Relative Date', and set 'Last 2 months' — this will ALWAYS show the last 2 months from today, no manual update needed

Filter Type 4 — Context Filter (Self Learning Topic)

A Context Filter is a special 'parent filter' that runs BEFORE all other filters. It narrows down the dataset first — then all other filters, calculations, and Top N filters run WITHIN that narrowed dataset.

Real-life example: *Imagine you want Top 3 Electronics products by sales. Without a Context Filter: Tableau first finds Top 3 products globally (Laptop, Mobile Phone, Tablet) and THEN checks if they are Electronics. Since they happen to all be Electronics, it works — but it might not always give the right result. With a Context Filter: Tableau first filters to Electronics ONLY, then finds Top 3 within that group. Now you are guaranteed to get the right answer — Top 3 from Electronics only.*

When MUST you use a Context Filter?

You MUST use a Context Filter when you combine a Category Filter with a Top N Filter. For example: 'Show Top 3 products, but only from the Accessories category.' Without Context Filter, you may get 0, 1, or 2 results instead of 3, because the Top N runs on all categories first.

How to create a Context Filter:

1	First, add your Category filter to the Filters shelf (e.g. Category = Electronics)
2	Right-click that filter pill on the Filters shelf
3	Click 'Add to Context'
4	The filter pill turns GREY — this means it is now a Context Filter (runs first)
5	Now add your Top N filter or other filters — they will run WITHIN the Electronics context
6	Result: You are guaranteed to get Top 3 of Electronics only, not Top 3 of everything

Topic 3 — Top N Filters

Top N Filter means showing only the TOP (or BOTTOM) N items from your data — the 5 best-selling products, the 3 most profitable cities, the 10 lowest-scoring students. It is one of the most useful features for business dashboards.

Real-life analogy

Forbes Rich List — instead of showing all billionaires in the world (thousands of people), they show the Top 100 by net worth. YouTube Trending shows the Top 50 videos. IPL Orange Cap shows the Top 10 run-scorers. All of these are 'Top N' lists.

3.1 Method 1 — Built-in Top N Filter (Quick Method)

Tableau has a built-in Top N option inside the Dimension Filter dialog. This is the quickest way to create a Top N filter — no formulas needed.

Example — Top 3 Products by Sales:

Rank	Product	Total Sales (₹)	Shown by Top 3 Filter?
1	Laptop	1,17,000	✓ YES — Rank 1
2	Mobile Phone	68,000	✓ YES — Rank 2
3	Tablet	47,000	✓ YES — Rank 3
4	Headphones	18,900	✗ NO — Rank 4, excluded
5	Keyboard	11,000	✗ NO — Rank 5, excluded

Step-by-step — Built-in Top N:

1	Build a bar chart: Product on Rows, SUM(Sales) on Columns
2	Drag 'Product' to the Filters shelf (drag the dimension, not the measure)
3	The Filter dialog opens — click the 'Top' tab at the top of the dialog
4	Select 'By Field' radio button
5	Set: Top → 3 → by → Sales → Sum (This means: show the Top 3 products ranked by Sum of Sales)
6	Click OK — your chart now shows only the 3 highest-selling products
7	Sort the bars: click the Sort Descending button — Laptop, Mobile Phone, Tablet in order

Top 3 Cities by Profit — another quick example:

1	Build chart: City on Rows, SUM(Profit) on Columns
2	Drag 'City' to the Filters shelf → click Top tab
3	Set: Top → 3 → by → Profit → Sum
4	Click OK
5	Result: Mumbai (₹25,200), Delhi (₹17,600), Bangalore (₹7,200) — only these 3 cities shown

Bottom N works the same way!

In the Top tab, change 'Top' to 'Bottom' to find your worst performers — lowest profit cities, slowest-moving products, lowest-scoring students. The setup is identical, just change the dropdown.

3.2 Method 2 — Dynamic Top N using a Parameter

The built-in method is fixed — if you set Top 3, it always shows 3. With a Parameter, the user can CHANGE the N value themselves by moving a slider. This makes your dashboard interactive and much more powerful.

Real-life example: *A manager wants to see Top 5 today, Top 10 tomorrow for a different meeting. Instead of you rebuilding the chart twice, a Parameter lets them drag a slider themselves. They control the N, you do not have to do anything.*

Full Step-by-Step — Dynamic Top N with Parameter

We will build: 'Top N Products by Sales' where the user controls N using a slider.

STEP 1 Create Parameter	In the Data panel, click the dropdown arrow → Create Parameter Name: Top N Products Data Type: Integer Current Value: 3 (default shown on load) Allowable Values: Range Minimum: 1 Maximum: 5 Step Size: 1 Click OK
STEP 2 Create Calculated Field	Go to Analysis → Create Calculated Field Name: Top N Rank Formula: <code>RANK(SUM([Sales (₹)]))</code> Click OK This calculates the sales rank of each product (1 = highest)

STEP 3 Build the Chart	Drag Product to Rows Drag SUM(Sales) to Columns Sort descending by Sales You should see all 5 products as bars
STEP 4 Add Top N Rank to Filters	Drag 'Top N Rank' (your calculated field) to the Filters shelf Tableau asks what type — choose 'All Values' In the filter dialog, select '≤' (less than or equal to) Type the value: [Top N Products] — reference your parameter! Or: use the formula <code>RANK(SUM([Sales (₹)])) ≤ [Top N Products]</code> directly
STEP 5 Alternative — Simpler Method	Instead of the filter dialog, right-click 'Top N Rank' on Filters shelf Choose 'Edit Filter' → use condition: Top N Rank ≤ [Top N Products] This is the cleanest approach
STEP 6 Show Parameter Control	Right-click 'Top N Products' in the Data panel Click 'Show Parameter Control' A slider (1 to 5) appears on the right side of your view
STEP 7 Test It!	Move the slider from 1 to 5 Slider at 1 → only Laptop shown Slider at 3 → Laptop, Mobile Phone, Tablet Slider at 5 → all 5 products Your dynamic Top N chart is complete!

What the results look like at different N values:

Parameter Value (N)	Products Shown	Total Sales Shown (₹)
N = 1	Laptop only	1,17,000
N = 2	Laptop, Mobile Phone	1,85,000
N = 3	Laptop, Mobile Phone, Tablet	2,32,000
N = 4	Laptop, Mobile Phone, Tablet, Headphones	2,50,900
N = 5	All 5 products (full view)	2,62,600

3.3 The Critical Context Filter Problem — and the Fix

This is the most common mistake beginners make with Top N filters. Read this carefully — it will save you a lot of confusion.

The Problem:

You want to see 'Top 3 Products from the Accessories category only'.

- You add a Dimension Filter: Category = Accessories
- You add a Top 3 filter: Top 3 by Sales
- You expect to see: *Headphones, Keyboard, and... one more Accessories product*
- **What you actually get:** Only Headphones and Keyboard (2 results, not 3) — because the Top 3 ran on ALL products first, gave Laptop, Mobile, Tablet... then the Category filter kept only accessories. Almost none of the Top 3 are accessories!

The Fix — Add Category filter as a Context Filter:

1	Add Category filter (Accessories) to the Filters shelf
2	Right-click the Category filter pill → 'Add to Context'
3	The pill turns GREY — it now runs FIRST before everything else
4	Now add the Top 3 filter on Product
5	Tableau first narrows dataset to Accessories only, THEN finds Top 3 in that group
6	Result: Headphones (₹18,900) and Keyboard (₹11,000) — only 2 Accessories products exist, so Top 3 naturally shows both. If you had 10 accessories, you would correctly see only the top 3.

The Golden Rule

Whenever you combine a Category/Dimension filter WITH a Top N filter, ALWAYS make the Category filter a Context Filter. Right-click → Add to Context. This one step fixes the most common Top N bug in Tableau.

3.4 Top N vs Bottom N — Quick Comparison

Top N Filter	Bottom N Filter
Shows the BEST N performers	Shows the WORST N performers
Sorted highest to lowest	Sorted lowest to highest
Use case: Top 5 sales reps	Use case: Bottom 3 cities by profit
Use case: Top 10 customers by revenue	Use case: 5 slowest-moving products
In filter dialog: select 'Top'	In filter dialog: select 'Bottom'

Module 4 — Quick Reference Cheat Sheet

Sorting — All 4 Types

Sort Type	When to Use	How to Do It in Tableau
Alphabetical	City names, product names — no natural ranking	Click the A→Z icon on the axis, or right-click pill → Sort → Alphabetic
Computed	Rank products/cities by Sales, Profit, Qty	Right-click pill → Sort → Field → choose measure + aggregation
Manual	Months, weekdays, process stages (custom order)	Right-click pill → Sort → Manual → drag items to desired order
Nested	Sort WITHIN groups (e.g. top product per category)	Right-click INNER dimension pill → Sort → check 'Sort within panes'

Filter Types — Summary

Filter Type	Filters On	Best Used For
Dimension	Text / Category fields	Show only Electronics, only Mumbai
Measure	Number fields	Show Sales > 20,000, Profit between 2k-10k
Date	Date / Time fields	Last 30 days, Q1 only, specific date range
Context	Runs before all others	Parent filter for Top N + Category combos
Table Calc	Post-calculation results	Filter on RANK, RUNNING_SUM after they compute

Top N Filter — Two Methods

Method 1: Built-in (Fixed N)	Method 2: Parameter (Dynamic N)
Drag dimension to Filters shelf	Create Parameter (Integer, Range 1-N)
Click 'Top' tab in filter dialog	Create Calculated Field: RANK(SUM([Sales]))
Set: Top 3 by Sales Sum	Add calc field to Filters, set <= [Parameter]
Click OK — always shows exactly 3	Right-click parameter → Show Parameter Control

Method 1: Built-in (Fixed N)	Method 2: Parameter (Dynamic N)
User cannot change the N value	User drags slider to change N anytime
Good for fixed reports	Good for interactive dashboards

The Context Filter Rule

Remember Always

Category Filter + Top N Filter together = You MUST make the Category filter a Context Filter.
 Right-click the Category filter pill on the Filters shelf → Add to Context → pill turns GREY.
 Without this: wrong results. With this: correct Top N within your selected category.

Filter Order of Operations (Quick Reference)

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RUNS FIRST → Extract Filter  (at data source level)
            → Data Source Filter  (limits data coming in)
            → Context Filter  (parent filter – grey pill)
            → Dimension Filter  (text/category fields)
            → Measure Filter  (numeric fields)
RUNS LAST  → Table Calculation Filter  (post-calculation)
  
```

Tip: Filters that run EARLIER affect all filters that come AFTER them.

End of Module 4 — Sorting, Filtering & Top N in Tableau

Practice Exercise: Using the 10-row dataset, build Top 3 Electronics products by Sales with a dynamic parameter slider!